Name _____

Comparison Problems

Essential Question How does a model help you solve a comparison problem?

Operations and Algebraic Thinking—4.OA.2

MATHEMATICAL PRACTICES
MP.1, MP.3, MP.4, MP.7

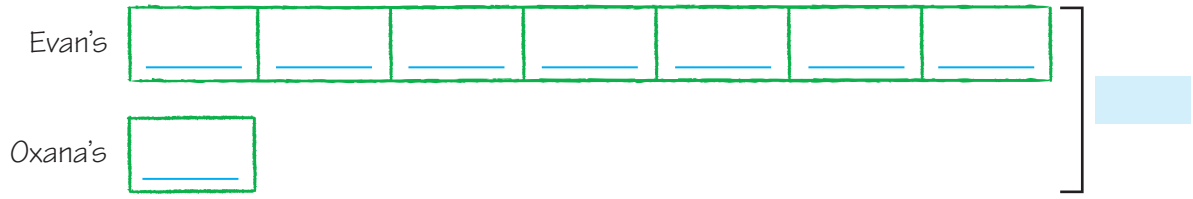
Unlock the Problem Real World

Evan's dog weighs 7 times as much as Oxana's dog. Together, the dogs weigh 72 pounds. How much does Evan's dog weigh?

Example 1 Use a multiplication model.

STEP 1 Draw a model. Let *n* represent the unknown.

Think: Let *n* represent how much Oxana's dog weighs. Together, the dogs weigh 72 pounds.



STEP 2 Use the model to write an equation. Find the value of *n*.

_____ × *n* = _____ **Think:** There are 8 parts. The parts together equal 72.

8 × _____ = 72 **Think:** What times 8 equals 72?

The value of *n* is 9.

n is how much _____ weighs.

STEP 3 Find how much Evan's dog weighs.

Think: Evan's dog weighs 7 times as much as Oxana's dog.

Evan's dog = _____ × _____ Multiply.

= _____

So, Evan's dog weighs 63 pounds.

Math Talk **Mathematical Practices**

Explain how you know you have found the weight of Evan's dog.

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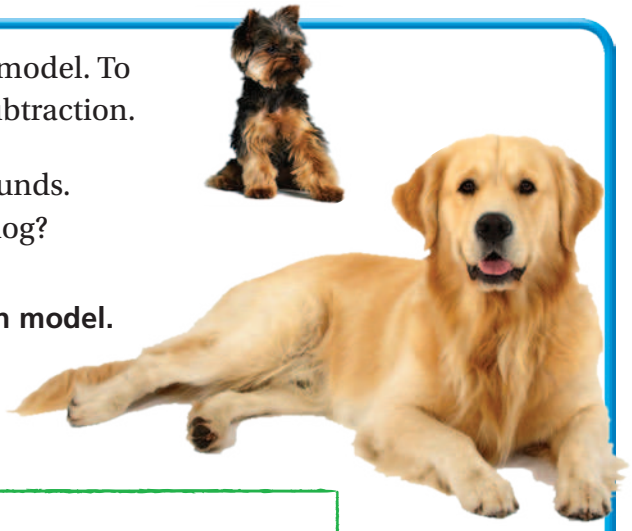
To find how many times as much, use a multiplication model. To find how many more or fewer, model the addition or subtraction.

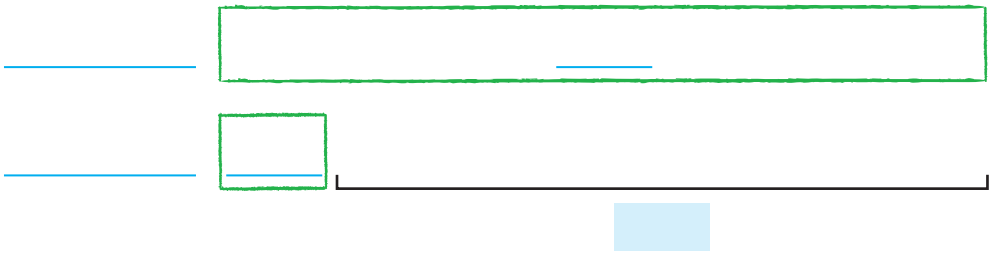


Evan's dog weighs 63 pounds. Oxana's dog weighs 9 pounds. How much more does Evan's dog weigh than Oxana's dog?

Example 2 Use an addition or subtraction model.

STEP 1 Draw a model. Let *n* represent the unknown.

Think: Let *n* represent the difference.



STEP 2 Use the model to write an equation. Find the value of *n*.

_____ − _____ = *n* **Think:** The model shows a difference.

63 − 9 = _____ Subtract.

The value of *n* is _____.

n is _____.

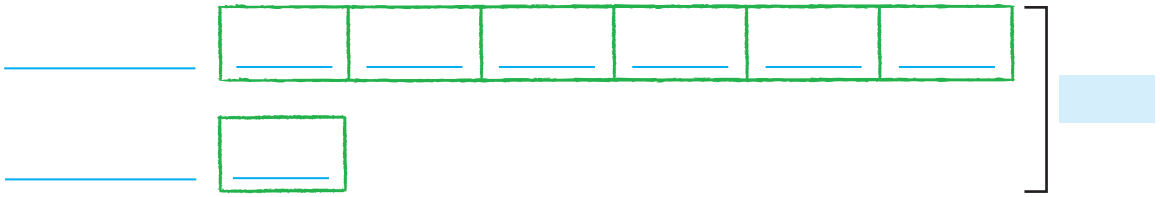
So, Evan's dog weighs 54 pounds more than Oxana's dog.

Share and Show

MATH BOARD

1. Maria's dog weighs 6 times as much as her rabbit. Together the pets weigh 56 pounds. What does Maria's dog weigh?

Draw a model. Let *n* represent the unknown.



Write an equation to find the value of *n*. $7 \times n =$ _____. *n* is _____ pounds.

Multiply to find how much Maria's dog weighs. $8 \times 6 =$ _____

So, Maria's dog weighs _____ pounds.

Math Talk **Mathematical Practices**

Explain how you can choose a model to help solve a comparison problem.

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Name _____

Draw a model. Write an equation and solve.

2. Last month Kim trained 3 times as many dogs as cats. If the total number of cats and dogs she trained last month is 28, how many cats did Kim train?

Draw a model.

Write an equation and solve.

3. How many more dogs than cats did Kim train?

Draw a model.

Write an equation and solve.

On Your Own

Practice: Copy and Solve **Draw a model.**

Write an equation and solve.

4. At the dog show, there are 4 times as many boxers as spaniels. If there are a total of 30 dogs, how many dogs are spaniels?

5. There are 5 times as many yellow labs as terriers in the dog park. If there are a total of 18 dogs, how many dogs are terriers?

6. Ben has 3 times as many guppies as goldfish. If he has a total of 20 fish, how many guppies does he have?

7. Carlita saw 5 times as many robins as cardinals while bird watching. She saw a total of 24 birds. How many more robins did she see than cardinals?

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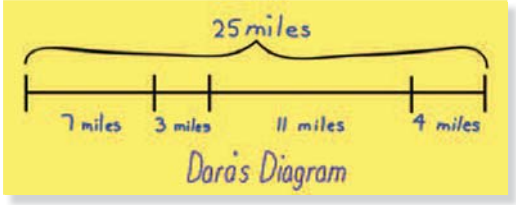
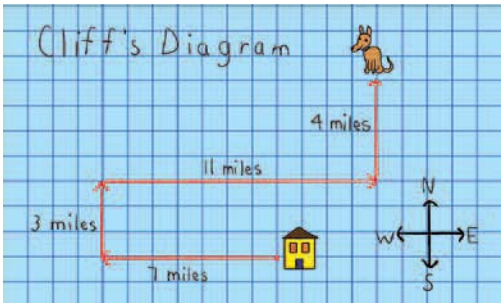
Problem Solving • Applications (Real World)

8. GO DEEPER To get to a dog show, Mr. Luna first drives 7 miles west from his home and then 3 miles north. Next, he turns east and drives 11 miles. Finally, he turns north and drives 4 miles to the dog show. How far north of Mr. Luna's home is the dog show?

To solve the problem, Dara and Cliff drew diagrams. Which diagram is correct? Explain.





_____

_____

_____

_____

9. MATHEMATICAL PRACTICE 2 **Use Reasoning** Valerie and Bret have a total of 24 dog show ribbons. Bret has twice as many ribbons as Valerie. How many ribbons does each have?

_____

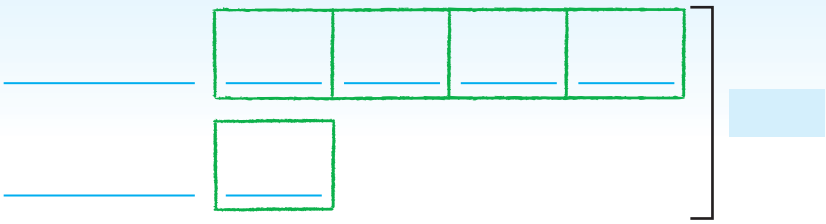
_____

WRITE Math
Show Your Work

10. THINK SMARTER Noah built a fenced dog run that is 8 yards long and 6 yards wide. He placed posts at every corner and every yard along the length and width of the run. How many posts did he use?

_____

11. THINK SMARTER Last weekend, Mandy collected 4 times as many shells as Cameron. Together, they collected 40 shells. How many shells did Mandy collect? Complete the bar model. Then write an equation and solve.



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Standards Practice Book